~~Porous PI/h-BN films made by NIPS over the full 0–70 wt% h-BN range 3-phase Lewis–Nielsen model unifies dielectric and thermal behavior Pore shape factor $A = 0.08$ fingerprints NIPS sponge vs. spherical pores Single-parameter thermal fit: $\lambda_{\mathrm{BN,eff}} = 6.8\,\mathrm{W\,m^{-1}\,K^{-1}}$, interface-limited 70 wt% h-BN locus is the global Pareto frontier of the system~~

~~Multifunctional~~ Model-Guided Design of Porous Polyimide/~~Hexagonal~~ Boron Nitride ~~Composite~~ Films ~~Fabricated by Non-Solvent Induced Phase Separation: Coupled Experimental Characterization and 3-Phase Lewis–Nielsen Composite Modeling~~for Low-Permittivity, Thermally Conductive Packaging

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Abstract

~~Polymer films for next-generation electronic packaging~~ Packaging films for high-frequency and power electronics must combine a low dielectric constant with ~~sufficient~~ useful thermal conductivity and dimensional stability~~.~~ ~~Most high-thermal-conductivity fillers are electrically conductive and destroy the dielectric character of the film, so ceramic fillers such as hexagonal boron nitride (h-BN) are preferred, although their affinity with polymer matrices is often limited~~, but the strategies that improve one property usually degrade another. Porous polymers ~~offer excellent low-permittivity dielectrics, but porosity suppresses thermal conductivity and mechanical~~ reach the lowest permittivities at the expense of heat transport and strength, and ~~high filler loadings are difficult to achieve in porous architectures. Here ,~~ thermally conductive fillers raise permittivity. Here we report porous polyimide (PI)/hexagonal boron nitride (h-BN~~composite filmsspanning 0–70 wt% h-BN were fabricated~~) films, made by non-solvent induced phase separation (NIPS) ~~, and~~ over 0–70 wt% h-BN and restructured by hot-pressing~~was applied to restructure the pore network, improving mechanical strength and raising thermal conductivity while retaining the low permittivity of the porous state. The ,~~ in which porosity and filler loading are treated as two separate design variables. We measure the dielectric, thermal, thermomechanical, and mechanical properties of all ten films ~~were measured, and the full dataset was then interpreted and verified within a single calibrated~~ and describe them with a single three-phase Lewis–Nielsen ~~framework treating each film as~~ model in which each film is a PI, h-BN, and air system. The fitted pore shape factor ($A_{\text{pore}} = 0.08$) fingerprints the bicontinuous NIPS sponge~~structure, and the framework ,~~, and a single-parameter thermal calibration reproduces the conductivity data to within $0.03\,\mathrm{W\,m^{-1}\,K^{-1}}$. The model identifies 70 wt% h-BN ~~with controlled~~ at 50–60% porosity as the ~~optimal design point, where the film achieves~~ most favorable design region: the non-pressed film reaches $\varepsilon_r = 1.28$ ~~,~~ and $\lambda = 0.31\,\mathrm{W\,m^{-1}\,K^{-1}}$, and ~~CTE = 10.1 ppm/K.~~ the hot-pressed film combines $\varepsilon_r = 1.62$ and $\lambda = 0.28\,\mathrm{W\,m^{-1}\,K^{-1}}$ with a CTE of $10.1\,\mathrm{ppm\,K^{-1}}$. These results establish the pore shape factor as a microstructural design parameter for porous ceramic-filled polymer films and provide a calibrated basis for selecting composition and porosity in NIPS-derived packaging films.

Keywords: Polymer-matrix composites, Hexagonal boron nitride, Porous film, Lewis–Nielsen model, Dielectric properties, Thermal conductivity

1. Introduction

~~The convergence of high-density~~ High-density microelectronics, millimeter-wave fifth-generation (5G) communication, and electric-vehicle power electronics ~~is driving a fundamental shift in the property requirements for polymer substrate and packaging materials. Contemporary device architectures demand films that~~

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~~simultaneously satisfy three stringent criteria:~~ place three demands on a polymer packaging film at once: a low dielectric constant ($\varepsilon_r < 2.5$) to ~~minimize~~ limit signal delay and cross-talk at GHz–THz ~~operating frequencies; adequate thermal conductivity ($\lambda > 0.2\,\mathrm{W\,m^{-1}\,K^{-1}}$)~~ frequencies; a thermal conductivity above $\sim 0.2\,\mathrm{W\,m^{-1}\,K^{-1}}$ to dissipate heat fluxes that ~~increasingly~~ now exceed $100\,\mathrm{W\,cm^{-2}}$ in advanced packages; and ~~dimensional stability, expressed as~~ a coefficient of thermal expansion (CTE~~< 20 ppm/K)~~) below $\sim 20\,\mathrm{ppm\,K^{-1}}$, matched to copper ~~metallization~~ ($\sim$~~17 ppm/K~~$17\,\mathrm{ppm\,K^{-1}}$) and silicon ~~substrates~~ ($\sim$~~3 ppm/K) to prevent fatigue delamination under thermal cycling~~ $3\,\mathrm{ppm\,K^{-1}}$) so that thermal cycling does not delaminate the stack [1, 2, 3]. No ~~single existing polymeric material satisfies all three criteria~~ existing polymer meets all three without modification, and the strategies ~~used to~~ that improve one property typically degrade another~~ — a fundamental challenge in multifunctional material~~. This is the central difficulty of multifunctional packaging design.

Aromatic polyimide (PI) ~~, synthesized from dianhydride and diamine monomers, has served as the foundational dielectric substrate material in the electronics industry for decades owing to its combination of high~~ has been the dielectric substrate of choice for decades because of its thermal stability (decomposition ~~temperature > 500 °C), outstanding mechanical properties~~above 500 °C), strength, chemical resistance, and electrical insulation [4]. The canonical PMDA–ODA system ~~employed in the present work exhibits~~ used here has a dielectric constant of $\sim$3.4–3.8, a thermal conductivity of $\sim 0.12\,\mathrm{W\,m^{-1}\,K^{-1}}$, and a CTE of $\sim$~~40–60 ppm/K for dense films — properties that increasingly fall short of 5G and power electronics requirements~~ $40\text{–}60\,\mathrm{ppm\,K^{-1}}$ in dense form, none of which meets the requirements above [5]. ~~Multiple molecular design strategies have been explored to intrinsically reduce~~ Molecular strategies that lower the dielectric constant~~of PI, including incorporation of fluorinated monomers(hexafluoropropylidene groups), introduction of~~, such as fluorinated monomers, bulky pendant groups~~that increase free volume, and use of~~, and low-polarizability ~~non-conjugated diamine structures [6, 7]. While these approaches can achieve $\varepsilon_r < 3.0$, they typically require specialized monomers , complex synthesis, and they do not address the thermal conductivitydeficit~~diamines, reach $\varepsilon_r < 3.0$ but need specialized monomers and do nothing for thermal conductivity [6, 7].

~~Porous structure engineering through~~ Porosity offers a complementary route. In non-solvent induced phase separation (NIPS)~~offers a distinct and complementary approach to dielectric reduction. In NIPS~~, a poly(amic acid) ~~/solvent (NMP) solution is cast onto a substrate~~ solution in NMP is cast and immersed in a non-solvent ~~bath, typically a lower-surface-energy solvent~~ such as acetone, water, or ~~alcohols. Rapid exchange of solvent and non-solvent across the film–bath interface drives thermodynamic instability —~~ an alcohol; rapid solvent exchange drives spinodal decomposition or ~~nucleation-and-growth — yielding a bicontinuous or sponge-like porous architecture that is preserved upon~~ nucleation and growth, and the resulting bicontinuous sponge survives thermal imidization [8, 9, 10]. Because air ~~($\varepsilon_{\mathrm{air}} = 1.0$)~~ has the lowest dielectric constant of any material, ~~incorporating large air volume fractions through NIPS can dramatically reduce the effective film permittivity far below any molecular modification strategy. Reported values span $\varepsilon_r = 1.51$–2.48 for~~ large air fractions lower the permittivity well below what molecular design alone achieves: NIPS-derived ~~porous~~ PI and related ~~engineering-polymer films with~~ engineering polymers reach $\varepsilon_r = 1.51$–2.48 at porosities up to 75%, ~~with the dielectric constant continuously tunable by adjusting~~ tunable through polymer concentration, non-solvent~~identity~~, and immersion conditions [7, 11, 12]. ~~The approach is also directly scalable and compatible with roll-to-roll processing, unlike aerogel or~~ Unlike aerogel and freeze-drying ~~methods that require supercritical drying or cryogenic equipment~~ routes, NIPS needs neither supercritical drying nor cryogenic equipment and is compatible with roll-to-roll processing [13].

~~The~~ Hexagonal boron nitride (h-BN) addresses the thermal and dimensional ~~limitations~~ shortfalls of porous PI~~can be addressed by incorporation of hexagonal boron nitride (h-BN) as a multifunctional inorganic filler. h-BN — also known as 'white graphite' due to its structural analogy with graphite —~~. It is a wide-bandgap ceramic (~~$E_g \approx 6$ eV) with a layered hexagonal crystal structure that confers exceptional~~ $E_g \approx 6$ eV) whose layered structure gives a strongly anisotropic thermal conductivity~~:~~, up to 400 $\mathrm{W\,m^{-1}\,K^{-1}}$ in-plane and $\sim$50 $\mathrm{W\,m^{-1}\,K^{-1}}$ through-plane, ~~placing it among the most thermally conductive electrically insulating materials known~~among the highest of any electrical insulator [14]. ~~The dielectric constant of h-BN~~ Its dielectric constant ($\sim$6.0; $\varepsilon_\perp = 7.04$, $\varepsilon_\parallel = 4.95$) is ~~substantially lower than competing thermally conductive ceramics such as~~ well below that of aluminum nitride ($\varepsilon_r \approx 9$) ~~and~~ or silicon carbide ($\varepsilon_r \approx 10$), ~~making it~~ so it is the filler of choice when dielectric performance must be preserved~~alongside thermal enhancement~~ [15]. ~~Furthermore, the~~ Its in-plane CTE ~~of h-BN~~ ($\sim$~~1.5 ppm/K) and its high in-plane elastic~~ 1.5 $\mathrm{ppm\,K^{-1}}$) and modulus ($\sim$~~200–700 GPa)~~ 200–700 GPa) also make it an effective dimensional stabilizer~~in polymer composites through elastic constraint mechanisms~~ [16, 17].

~~Prior studies of dense~~ Dense PI/h-BN ~~composite films — where the PI matrix is not porous — have demonstrated thermal conductivity values of~~ films reach 0.5~~–6.2 W/m K at BN loadings of~~–6.2 $\mathrm{W\,m^{-1}\,K^{-1}}$ at 10–60 wt% ~~, typically at the cost of increased dielectric constant (~~h-BN, but at $\varepsilon_r = 3.5$–5.0~~)~~ [18, 19, 20]. Yang et al. [21] ~~fabricated porous BN/PI composite films by combining~~ combined a high-internal-phase Pickering emulsion (HIPPE) ~~and~~ template with hot-pressing ~~, achieving simultaneously~~ to obtain porous h-BN/PI films with low ε_r and high thermal diffusivity. ~~However, the spherical~~The spherical, emulsion-templated pores ~~in~~ of HIPPE films are~~geometrically distinct~~, however, geometrically different from the bicontinuous sponge ~~pores generated by NIPS, resulting in fundamentally different pore-structure contributions~~ of NIPS, and the two architectures should contribute differently to composite properties~~—a distinction that has not previously been quantified within a unified modeling framework~~; this difference has not been quantified in a single model. Porous PI films ~~have likewise shown enhanced dielectric stability~~ also retain their dielectric properties at elevated temperature [22], ~~while~~ and electrospinning–hot-press routes ~~have produced dense BN~~give dense h-BN/PI composites ~~with outstanding~~ of high thermal stability [15]. ~~Missing from the literature~~ What is missing is a quantitative model that relates dielectric constant and thermal conductivity ~~simultaneously to both BN~~ to both h-BN loading and porosity, ~~provides physically interpretable composite~~ returns physically interpretable microstructure parameters, and ~~generates actionable~~ yields design guidance for ~~property optimization in~~ NIPS-derived porous PI/~~BN systems specifically fabricated by NIPS~~h-BN films.

The Lewis–Nielsen ~~model, originally formulated by Lewis and Nielsen [23, 24]as~~ (LN) model [23, 24], a modification of the Halpin–Tsai equations, provides ~~exactly~~ this capability. It predicts the effective scalar transport property of a ~~two-phase particulate composite as a function of~~ particulate composite from the filler volume fraction, the filler–matrix property contrast, ~~filler shape (through an~~ a filler shape factor (the Einstein coefficient A), and a maximum packing fraction $\phi_{\max}$. ~~By applying the model sequentiallyto a three-component system —~~ Applied sequentially, first mixing PI ~~matrix~~ with h-BN ~~filler to obtain a dense composite,~~ and then introducing air pores~~— a 3-phase LN model is constructed that uses~~, it becomes a three-phase model that takes measured porosities and ~~BN~~ loadings as inputs and ~~yields fitted parameters that directly~~ returns parameters that characterize the h-BN ~~nanoplatelet~~ platelet geometry and the ~~nature of the NIPS-generated~~ NIPS pore architecture [25]. ~~Analogously, the~~ The Turner model [26] ~~handles CTEthrough elastic-modulus-weighted mixing~~plays the corresponding role for CTE, and the Gibson–Ashby ~~model [27] provides the standard porosity-scaling law for mechanical properties~~ scaling [27] for the strength of open-cell porous solids.

~~In this work, we present the first systematic~~ Here we report a combined experimental and modeling study of NIPS-derived porous PI/h-BN ~~composite films across the full composition range (~~films across 0–70 wt% h-BN ~~)~~ in both non-pressed (NP) and hot-pressed (HP) states. The central contribution is ~~the calibrated 3-phase Lewis–Nielsen framework that integrates all measured properties — dielectric constant , thermal conductivity , CTE, and tensile strength — into a unified quantitative description, extracts physically meaningful composite microstructure~~ a calibrated three-phase LN framework that describes the measured dielectric constant and thermal conductivity with one shared microstructural parameterization, extracts interpretable composite parameters, and generates ~~2D design maps , an Ashby-style property diagram, and a global Pareto-frontier analysis to guide experimental optimization. Critically, the~~ design maps for simultaneous property optimization. The fitted pore shape factor $A_{\mathrm{pore}} = 0.08$ ~~provides a quantitative~~ is a microstructure fingerprint that distinguishes NIPS sponge ~~pore geometry~~ pores from the spherical pores of ~~HIPPE-based composites, the~~ HIPPE composites; a deliberately parsimonious single-parameter thermal calibration avoids the over-fitting ~~pitfalls~~ common to multi-parameter composite models~~, and the model-guided analysis directly~~; and the calibrated model prescribes the porosity ~~required to satisfy demanding~~ needed to meet dual-property specifications. We evaluate the framework against ten films to show that porosity and pore geometry, not filler geometry, govern the dielectric response, whereas thermal conductivity is gated by filler loading through the approach to maximum packing.

2. ~~Experimental~~Materials and methods

2.1. Materials

Pyromellitic dianhydride (PMDA, >98%, Tokyo Chemical Industry, Japan), 4,4′-oxydianiline (ODA, >98%, TCI), and hexagonal boron nitride (h-BN, 99.5%, mean particle diameter ∼70 nm, MK Impex Corp., Canada) were used as received~~without further purification~~. N-Methyl-2-pyrrolidone (NMP, 99.5%) and acetone (99.5%) were purchased from Duksan Pure Chemicals, Korea.

2.2. PAA/h-BN ~~Dispersion Synthesis~~dispersion synthesis

ODA (10 mmol, 2.0024 g) was dissolved in NMP (23.707 g~~, 15 wt% total solids~~) at room temperature~~. h-BN (0, 10, 30, 50, or 70 wt% relative to total PAA+BN mass) was dispersed~~ and h-BN was dispersed in the solution by probe sonication (30 min)~~to minimize agglomeration~~. PMDA (10 mmol, 2.1812 g) was added portionwise with ~~continuous stirring~~ stirring for 24 h at room temperature ~~for 24 h, yielding a viscous PAA~~to give a viscous poly(amic acid) (PAA)/h-BN ~~composite~~ solution at equimolar PMDA:ODA~~ratio. [editorial comment removed]~~. h-BN was added at 0, 10, 30, 50, or 70 wt% of the total PAA + h-BN solids while the PAA concentration in NMP was held at 15 wt% for all compositions, so that the phase-separation thermodynamics of the polymer solution were the same across the series and h-BN loading was the only compositional variable.

2.3. NIPS ~~Film Fabrication~~ film fabrication and ~~Thermal Imidization~~thermal imidization

The PAA/h-BN solutions were spin-coated onto soda-lime glass ~~substrates~~ (1000 rpm, 10 s)~~. [editorial comment removed]~~ ~~Films were immediately immersed in a 20 mL acetone non-solvent bath~~; the high viscosity of the 15 wt% PAA solution and the short spin time leave a wet film of several hundred micrometres, which after NIPS and imidization gives free-standing porous films of 52–172 µm (Figure 1). The coated substrates were immersed immediately in 20 mL of acetone at room temperature for 2 h~~to induce phase separation~~. The solidified ~~PAA/h-BN green films were thermally~~ films were imidized at 200 °C for 2 h ~~followed by~~ and 350 °C for 1 h ~~under ambient atmosphere, then~~ in air and peeled from the glass to ~~yield free-standing porous films designated NP (Non-Pressed) . Complete imidization was confirmed by~~ give the non-pressed (NP) films. FT-IR ~~: characteristic~~ (Figure S8) confirmed complete imidization: imide C=O stretching at ~~1780 cm^{-1} (asymmetric) and 1720 cm^{-1}(symmetric)~~1780 and 1720 cm^{-1}, C–N stretching at 1380 cm^{-1}, and ~~complete disappearance of the PAA~~ no residual amide carbonyl band at ∼1660 cm^{-1}.

2.4. ~~Hot-Press Densification~~Hot-press densification

A ~~subset~~ second set of NP films was densified ~~by uniaxial hot-pressing~~ between polished steel plates in a hydraulic press ~~to produce HP (Hot-Pressed)films, adapting the hot-pressing protocol established in our previous~~ (250 °C, 400 MPa, 30 min), following the protocol of our earlier work on NIPS-derived porous PI ~~composite films [28]: temperature 250 °C, applied pressure 400 MPa, dwell time 30 min[editorial comment removed]. Hot-pressing partially collapses the NIPS pore network, increasing bulk density and tensile strength while reducing porosity. Since the pressing temperature (250 °C) lies far~~ composites [28], to give the hot-pressed (HP) films. Because 250 °C lies well below the glass transition ~~temperature~~ of PMDA–ODA PI (> 400 °C [28]), densification proceeds by ~~thermally-assisted plastic collapse (buckling and yielding)~~ thermally assisted plastic collapse of the glassy pore walls rather than by viscous flow~~above T_g~~. This is consistent with the partial, rather than complete, ~~porosity reduction observed~~ loss of porosity and with the resistance of the rigid h-BN network to compaction ~~discussed in~~ (Section 4.1). NP and HP films ~~at identical~~ of the same composition were characterized ~~side-by-side for all properties~~side by side.

2.5. *Characterization*

Tensile ~~properties were measured by~~ strength was measured on a universal testing machine (~~UTM, 5 mm min^{-1}crosshead speed; rectangular specimens: width 10 mm~~ 5 mm min^{-1}; 10 mm wide, 50 mm gauge length, ~~gauge length 50 mm, thickness measured~~ thickness by micrometer; ASTM D882). Dielectric constant was measured at 100 kHz ~~by~~ with an LCR meter (~~in-house); in addition, an independent verification measurement of the~~ [model, manufacturer]; parallel-plate contact electrodes, [diameter] mm) on films [with/without] sputtered electrodes. The HP 30 wt% film was ~~performed~~ also measured independently at KOPTRI (Korea Polymer Testing & Research Institute~~) according to~~; ASTM D150~~(~~, 100 kHz to 13 MHz~~; Agilent LCR;~~ , Agilent LCR, G10 electrode~~; 23(2) °C, 45(5) %~~ , 23 ± 2 °C, 45 ± 5% RH); the ~~inter-laboratory comparison is discussed in Sections 3.4 and~~ two laboratories are compared in Section 4.2. Thermal diffusivity (α, laser flash) and specific heat ~~capacity~~ (C_p, DSC, ~~[editorial comment removed]; 5 °C min^{-1};~~ ASTM E1269, 5 °C min^{-1}, 0–300 °C) were measured at Korea University of Technology and Education (KOREATECH); bulk density (ρ) was ~~determined~~ obtained from film mass and optically measured dimensions~~; thermal conductivity was calculated as~~ , and $\lambda = \alpha\rho C_p$. CTE was measured by TMA (Q400 EM, TA Instruments; tension mode~~; 10 °C min^{-1}; N_2 100 mL min^{-1}; preload 0.05 N;~~ , 10 °C min^{-1}, N_2 100 mL min^{-1}, preload 0.05 N, 100–200 °C~~range~~; Kyung Hee University). ~~Porosity was determined by mercury porosimetry~~ Open porosity was measured by mercury intrusion (PM33GT, Quantachrome; Cooperative Equipment Center~~). [editorial comment removed] Microstructure and elemental distribution~~ , [institution]), and total porosity was calculated as $\phi_{\text{total}} = 1 - \rho/\rho_{\text{solid}}$ with ρ_{solid} from the PI and h-BN densities and the solid-phase composition (Section 3.1); the two are compared in Section 4.1. Microstructure and composition were examined by FE-SEM/EDS (JEOL-7800F and JEOL-7001F)~~.~~ , and FT-IR spectra were recorded on a Jasco FT/IR-460 Plus ~~spectrometer (Jasco, Tokyo, Japan;~~ (600–4000 cm^{-1}).

Unless stated otherwise, each property was measured on one specimen per composition and processing state ($n = 1$), so the reported values carry the measurement uncertainty only. Representative uncertainties are ±5% in α, ±3% in C_p, and ±8% in ρ (limited by thickness), which propagate to about ±10% in λ; ±[value] in ε_r; and ±[value] ppm K^{-1} in CTE. These bounds set how far individual residuals can be interpreted in Sections 4.2 and 4.3.

3. Modeling ~~Framework~~framework

3.1. *Sequential ~~3-Phase~~ three-phase Lewis–Nielsen ~~Model~~model*

~~Each PI/h-BN porous film is modeled~~ We treat each film as a three-component system ~~:~~ of PI matrix (property k_m), h-BN filler (~~property~~ k_f), and air pores (~~property~~ $k_{\text{pore}} \approx k_{\text{air}}$). ~~We note that the~~ The term "three-phase Lewis–Nielsen model" has ~~previously been used~~ been used before for a matrix–filler–interphase construction ~~,~~ in which the third phase is the interfacial polymer layer ~~surrounding~~ around the nanofiller [29]; ~~the present formulation is distinct, with~~ here the third phase ~~being the NIPS-generated air~~ is the NIPS pore network. ~~The sequential mixing strategy [25]proceeds in two steps. In Step~~Following the sequential mixing approach [25], step 1 ~~, the PI matrix~~ combines PI and h-BN ~~filler are combined to yield the dense (non-porous) compositeproperty k_{dense}:~~ into a dense composite,

$$\frac{k_{\mathrm{dense}}}{k_m} = \frac{1 + A_{\mathrm{BN}}\, B\, \phi_{\mathrm{BN}}}{1 - B\, \psi\, \phi_{\mathrm{BN}}}, \tag{1}$$

where

$$\cancel{B = \frac{k_f/k_m - 1}{k_f/k_m + A_{\mathrm{BN}}}, \qquad \psi = 1 + \frac{1-\phi_{\max}}{\phi_{\max}^2}\, \phi_{\mathrm{BN}}.}$$

$$B = \frac{k_f/k_m - 1}{k_f/k_m + A_{\mathrm{BN}}}, \qquad \psi = 1 + \frac{1-\phi_{\max}}{\phi_{\max}^2}\, \phi_{\mathrm{BN}}, \tag{2}$$

~~In Step~~and step 2 ~~, air pores are introduced into the dense~~ introduces the pores into that composite:

$$\frac{k_{\mathrm{final}}}{k_{\mathrm{dense}}} = \frac{1 + A_{\mathrm{pore}}\, B_p\, \phi_p}{1 - B_p\, \psi_p\, \phi_p}. \tag{3}$$

~~Although the Lewis–Nielsen model was originally~~ The LN model was formulated for elastic moduli and thermal conductivity [23, 24], ~~its application~~ but it applies without modification to the dielectric constant~~is formally exact~~: permittivity and thermal conductivity are ~~both~~ scalar transport properties governed by the same Laplace-type ~~field~~ equation, $\nabla \cdot (k \nabla u) = 0$, so ~~all~~ every effective-medium ~~formalisms transfer between them under the substitution $k \leftrightarrow \varepsilon_r$. Indeed, for spherical inclusions in~~ result transfers under $k \leftrightarrow \varepsilon$. In the dilute limit for spheres the LN expression reduces to ~~the~~ Maxwell–Garnett~~form~~, of which it is a generalization through ~~the shape factor~~ A and ~~the packing term~~ ψ. ~~The decisive advantage of LN~~ Its advantage over the standard dielectric mixing rules (Lichtenecker, Maxwell–Garnett, Bruggeman) for the present system is twofold: it describes ~~both~~ ε_r and λ with one shared microstructural parameterization, and it is the only ~~formalism among these that carries~~ one of these formalisms with an explicit pore-geometry parameter (A_{pore}) and a packing-limit divergence~~—~~, the two features on which ~~the central conclusions of this work~~ our conclusions rest. A ~~quantitative benchmark (Supplementary~~ benchmark against the zero-parameter mixing rules (Supplementary Material, Section S3.5) confirms that the calibrated LN model outperforms all ~~standard zero-parameter mixing rules, with~~ of them, by the largest margin (RMSE 0.26 vs. 0.55–0.90 ~~) in the moderate-porosity~~ over the 30–70 wt% ~~range~~films) at moderate porosity.

The same ~~formalism applies identically to dielectric constant~~ equations are used for ε_r and ~~thermal conductivity~~ λ. The ~~weight fraction of h-BN~~ h-BN weight fraction was converted to a volume fraction in the solid (PI + ~~BN) phase using~~ h-BN) phase with $\rho_{\mathrm{PI}} = 1.42\,\mathrm{g\,cm^{-3}}$ and $\rho_{\mathrm{BN}} = 2.29\,\mathrm{g\,cm^{-3}}$~~. Measured open porosity from mercury porosimetry was used directly~~, and the measured open porosity was used as ϕ_p

in Eq. 3~~. Full~~ ; the relation between open and total porosity is examined in Sections 4.1 and 4.2. The full derivation, parameter tables, and ~~complete~~ Python implementation are ~~provided in Supplementary Material~~ ~~(~~given in Sections S2–S5~~)~~.

3.2. Physical ~~Basis~~ basis of the Einstein ~~Coefficient~~ coefficient A

The Einstein coefficient A is the central microstructure parameter ~~in the Lewis–Nielsen model, originating from~~ of the model. It originates in the classical treatment of field or flux distortion around a single inclusion in an infinite matrix [23, 24]~~. Two distinct A values are required in the present~~ , and the three-phase system needs two of them: A_{BN} for the h-BN filler and A_{pore} for the ~~air~~ pore network.

For ~~platelet fillers, the theoretical~~ platelets, A depends on the aspect ratio $\alpha = D/t$ and on orientation relative to the ~~applied field. For the~~ field. The nominal aspect ratio of the 70 nm h-BN ~~particles used here (~70 nm mean diameter), the nominal bulk aspect ratio~~ would give $A_{\mathrm{rand}} \sim 4$–14~~. However, three physical factors systematically~~ , but three factors reduce the effective ~~aspect ratio~~value: (i) probe sonication ~~introduces edge fragmentation and partial delamination, reducing effective~~ fragments edges and partially delaminates the platelets, lowering α toward 5–7; (ii) 70 nm ~~BN particles exhibit a high tendency to form face-to-face stacks, creating more equiaxed effective~~ particles readily stack face to face into more equiaxed inclusions; and (iii) ~~rapid NIPS solvent exchange freezes~~ the rapid NIPS exchange freezes the particles in random orientations~~, preventing in-plane alignment~~. For randomly oriented platelets of ~~effective~~ $\alpha \approx 5$–7, $A_{\mathrm{rand}} \approx 4$–6 and $\phi_{\max} \approx 0.55$–0.65 [23, 24, 25]; ~~these values are adopted~~ we adopt these values for the thermal model ($A = 5.5$, $\phi_{\max} = 0.60$~~fixed~~; Section 3.4). ~~For the dielectric model, the fitted~~ The dielectric fit returns $A_{\mathrm{BN}} = 0.68$~~falls~~ , below the isolated-sphere value ~~($A = 1.5$); however,~~ of 1.5, but the sensitivity analysis (Section 4.6, Figure S1a) ~~demonstrates~~ shows that ε_r ~~predictions are~~ is nearly insensitive to A_{BN} ~~across~~ over 0.3–6.0~~—the pore network (Step~~: the pore term (step 2, Eq. 3) dominates the dielectric response, so the dielectric A_{BN} is ~~only weakly determined by the data~~ weakly determined and its precise value carries little physical weight.

For the ~~NIPS-generated pore network~~pores, A ~~captures~~ measures how effectively the ~~dispersed~~ pore phase shields the matrix from the ~~applied field. For isolated spherical pores ,~~ field. Isolated spherical pores give $A = 1.5$. ~~For the bicontinuous sponge morphology produced by NIPS~~ In the bicontinuous NIPS sponge, the PI skeleton ~~forms~~ provides continuous conduction paths through the pore space~~—meaning~~ , so the field disruption per unit porosity is ~~greatly reduced~~much smaller. The fitted $A_{\mathrm{pore}} = 0.08$ quantifies this~~geometry~~: at $\phi_p = 0.49$ (NP 30 wt%~~film), isolated spherical pores ($A = 1.5$) would predict~~), spherical pores would give $\varepsilon_r \approx 2.22$ ~~, while~~ whereas $A_{\mathrm{pore}} = 0.08$ ~~predicts $\varepsilon_r \approx 1.71$—a~~ gives 1.71, and the difference of 0.51 ~~units that directly reflects the bicontinuous~~ is the signature of the sponge architecture. ~~This~~ A_{pore} ~~value constitutes a quantitative microstructure fingerprint distinguishing~~ therefore serves as a microstructure fingerprint that separates NIPS sponge pores from ~~spherical Pickering emulsion~~ the spherical Pickering-emulsion (HIPPE) pores of Yang et al. [21], for which $A \sim 1.2$–1.5 ~~would be~~ is expected.

3.3. CTE and ~~Mechanical Models~~mechanical models

CTE was predicted ~~using~~ with the Turner model [26]~~: —~~, $\mathrm{CTE}_c = (\sum_i \phi_i E_i \alpha_i)/(\sum_i \phi_i E_i)$, ~~with~~ using $E_{\mathrm{BN}} = 200\,\mathrm{GPa}$ and ~~$\alpha_{\mathrm{BN}} = 1.5$ ppm/K [16]. For α_{PI}, the measured neat PMDA–ODA porous PI value (HP sample #1~~$\alpha_{\mathrm{BN}} = 1.5\,\mathrm{ppm\,K^{-1}}$ [16], $E_{\mathrm{PI}} = 3.5\,\mathrm{GPa}$ [5], and for α_{PI} the value measured on the neat porous PI HP film ($41.92\,\mathrm{ppm\,K^{-1}}$, TMA) ~~of 41.92 ppm/K was used —~~rather than the ~~commonly cited dense PI~~ dense-PI literature value of $\sim$~~60 ppm/K — because the porous sponge architecture modifies the effective thermal~~ $60\,\mathrm{ppm\,K^{-1}}$, because the sponge architecture changes the effective expansion of the PI phase. ~~$E_{\mathrm{PI}} = 3.5$ GPa was taken from literature for PMDA–ODA PI [5]. For comparison, the linear rule-of-mixtures~~ The linear rule of mixtures (ROM) ~~prediction is also shown . Notably~~is shown for comparison. As discussed in Section 4.4, the Turner model ~~substantially~~ underpredicts and the ROM overpredicts the ~~experimental~~ measured CTE at 10–70 wt% ~~BN. This is attributed to the porous sponge architecture violating~~ h-BN because the porous skeleton violates the continuous load-path assumption of the Turner model: ~~h-BN~~ platelets embedded in discontinuous thin PI ligaments cannot ~~exert the same macroscopic elastic constraint as in a fully~~ constrain the matrix as they would in a dense co-continuous composite~~. The experimental CTE values therefore lie between the upper bound (ROM, no constraint) and the lower bound (Turner, full constraint), a physically reasonable outcome for a partially load-bearing porous skeleton~~, so the measured values fall between the two bounds. Tensile strength was modeled ~~via~~ with the LN composite modulus ~~prediction combined with~~ and the Gibson–Ashby open-cell ~~foam scaling(~~scaling, $\sigma \propto E_{\mathrm{dense}}(1-\phi_p)^2$~~)~~ [27].

3.4. Parameter ~~Estimation Protocol~~ estimation and ~~Model Validation~~model validation

Dielectric model~~.~~. ~~Four~~ We fitted four parameters (A_{BN}, $\phi_{\max,\mathrm{BN}}$, A_{pore}, $\phi_{\max,\mathrm{pore}}$) ~~were simultaneously optimized to minimize the sum of squared residuals over all 10~~ to the ten ε_r measurements (~~5~~ five compositions × ~~2 conditions) using~~ two states) by multi-start Nelder–Mead ~~optimization~~ minimization of the squared residuals (16 ~~initial conditions~~starts; Eq. S8). ~~The~~ Both packing parameters converge to the upper ~~boundary~~ bound (0.99) and are effectively inactive: ~~a sensitivity scan (Supplementary Table S11) shows that~~ fixing $\phi_{\max,\mathrm{BN}}$ anywhere in 0.50–0.99 changes the RMSE by only 0.03 (Table S11). The dielectric model is therefore governed by ~~2–3~~ two to three active parameters fitted to ~~10~~ ten points.

Thermal model~~.~~. When all three thermal parameters ($\lambda_{\mathrm{BN,eff}}$, $A_{\mathrm{BN},\lambda}$, $\phi_{\max,\mathrm{BN},\lambda}$) are fitted ~~without constraints~~freely, the optimizer converges to a degenerate solution ($A \to 700$, $\phi_{\max} \to 0.10$)~~because~~: A and $\phi_{\max}$ are strongly correlated in Eq. 1~~and~~, the data cannot separate them~~; the resulting parameters are physically meaningless even though the fit quality is good (Supplementary~~, and the fit is good but meaningless (Section S4.2). ~~To eliminate this degeneracy, the thermal model was deliberately reduced to a single fitted parameter: $A_{\mathrm{BN},\lambda}$ was fixed at 5.5 and $\phi_{\max,\mathrm{BN},\lambda}$ at 0.60 —~~We therefore fixed $A_{\mathrm{BN},\lambda} = 5.5$ and $\phi_{\max,\mathrm{BN},\lambda} = 0.60$, the literature values for randomly oriented platelets ~~with effective aspect ratio~~ of $\alpha \approx 5$–7 [23, 24, 25] ~~,~~ and consistent with the sonication-modified particle geometry ~~discussed in~~ of Section 3.2~~— and~~, and fitted only ~~was fitted , yielding 6.81 W m^{-1} K^{-1} with RMSE (~~$\lambda_{\mathrm{BN,eff}}$. The result, $\lambda_{\mathrm{BN,eff}} = 6.81\,\mathrm{W\,m^{-1}\,K^{-1}}$ with RMSE 0.027 ~~NP,~~

~~0.030 HPW/m K)~~(NP) and 0.030 W m^{-1} K^{-1} (HP), is statistically indistinguishable from the unconstrained ~~three-parameter~~ fit (0.027, 0.023)~~. The thermal model thus has~~, and every reported parameter is physically interpretable at a 7:1 data-to-parameter ratio~~, and all reported parameters are physically interpretable. Notably, with~~. With $\phi_{\max} = 0.60$, the 70 wt% composition ($\phi_{\mathrm{BN}} = 0.591$) sits at 99% of maximum packing, so the ~~packing-interaction~~ packing term ψ in Eq. 1 ~~naturally generates~~ produces the sharp conductivity ~~upturn observed experimentally~~ rise seen at this loading~~—~~; this is the LN representation of ~~approaching the particle-contact (percolation) limit~~the approach to particle contact.

Validation~~.~~. ~~An independent KOPTRI measurement of the HP 30 wt% film gives $\varepsilon_r = 1.610$, compared with 1.864 in-house (−13.6%~~ The calibrated dielectric model is compared with an independent inter-laboratory ~~deviation) and a model prediction of 1.376 at the measured porosity of 67.2%. All three values disagree pairwise at the 14% level — and all of the disagreement traces to the same composition flagged for the porosity anomaly~~ measurement in Section ~~4.1: if the HP 30 wt% porosity lay in the 40–45% range expected from the monotonic NP→HP densification trend (rather than the anomalous measured 67.2%), the model would predict $\varepsilon_r = 1.79$–1.92, in agreement with the in-house measurement. The inter-laboratory comparisontherefore reinforces, rather than undermines, the conclusion that the 30 wt% HP porosity measurement requires re-examination. Model credibility instead~~ 4.2. Beyond that comparison, its credibility rests on three ~~pillars~~points: (i) the parsimony of the calibration (~~2–3~~ two to three active dielectric parameters, ~~1~~ one thermal parameter); (ii) the shape-factor sensitivity analysis (Section 4.6)~~demonstrating that conclusions are robust to~~, which shows that the conclusions do not depend on the weakly determined parameters; and (iii) the agreement of the model λ envelope with the ~~experimental~~ measured 70 wt% ~~data~~ films (Section 4.8, Figure ~~??e~~4d).

Table 1: ~~3-phase~~ Three-phase Lewis–Nielsen parameters: estimation protocol and physical interpretation.

Parameter	Symbol	Value	Status	Physical interpretation
h-BN shape factor (ε_r)	A_{BN}	0.68	Fitted (weakly det.)	ε_r is pore-dominated; A_{BN} carries little weight (Figure S1a)
h-BN max. packing (ε_r)	$\phi_{\max,\mathrm{BN}}$	0.99	Boundary (inactive)	RMSE changes ≤ 0.03 over 0.50–0.99 (Table S11)
Pore shape factor	A_{pore}	0.08	Fitted	Bicontinuous NIPS sponge; cf. $A = 1.5$ for isolated spheres
Pore max. packing	$\phi_{\max,\mathrm{pore}}$	0.99	Boundary	NIPS sponge supports up to 91% open porosity
h-BN shape factor (λ)	$A_{\mathrm{BN},\lambda}$	5.5	Fixed (literature)	Randomly oriented platelet, effective $\alpha \approx 5$–7
h-BN max. packing (λ)	$\phi_{\max,\mathrm{BN},\lambda}$	0.60	Fixed (literature)	Random platelet packing; $\phi_{\mathrm{BN}}(70\,\mathrm{wt\%}) = 0.59 \rightarrow 99\%$ of $\phi_{\max}$
Eff. ~~BN~~ h-BN conductivity	$\lambda_{\mathrm{BN,eff}}$	~~6.81 W/m K~~ 6.81 W m^{-1} K^{-1}	Fitted (single)	~~≈ 7× below bulk (~50)~~Well below the bulk through-plane value; Kapitza resistance at 70 nm

4. Results and ~~Discussion~~discussion

4.1. Film ~~Morphology:~~ morphology in the NP ~~vs.~~ and HP ~~Processing States~~states

~~Two distinct film states result from NIPS fabrication and optional hot-press densification. Non-Pressed (NP) films retain the full porous sponge architecture generated during NIPS : acetone non-solvent exchange produces a bicontinuous pore network with characteristic pore sizes of~~ NIPS with acetone gives a bicontinuous sponge with 1–10 µm ~~confirmed by FE-SEM. Measured~~ pores in every NP film, with the h-BN platelets embedded in the PI cell walls (Figure 1; photographs and top-view micrographs in Figures S6 and S7). Open porosities range from 48.8% (30 wt%~~h-BN~~) to 91.4% (10 wt%~~h-BN~~). The ~~anomalously high porosity~~ high value at 10 wt% ~~BN is attributed to~~ may reflect a nucleation effect~~:~~, with the h-BN ~~nanoparticles at this loading may act~~ particles acting as additional phase-separation nuclei ~~, producing finerand~~ that give finer, more uniformly distributed pores. ~~FE-SEM cross-sections confirm the sponge-like open-cell architecture in all NP films, with h-BN platelets embedded within the PI cell walls. Figure 1 presents the cross-sectional morphology of all ten films.~~

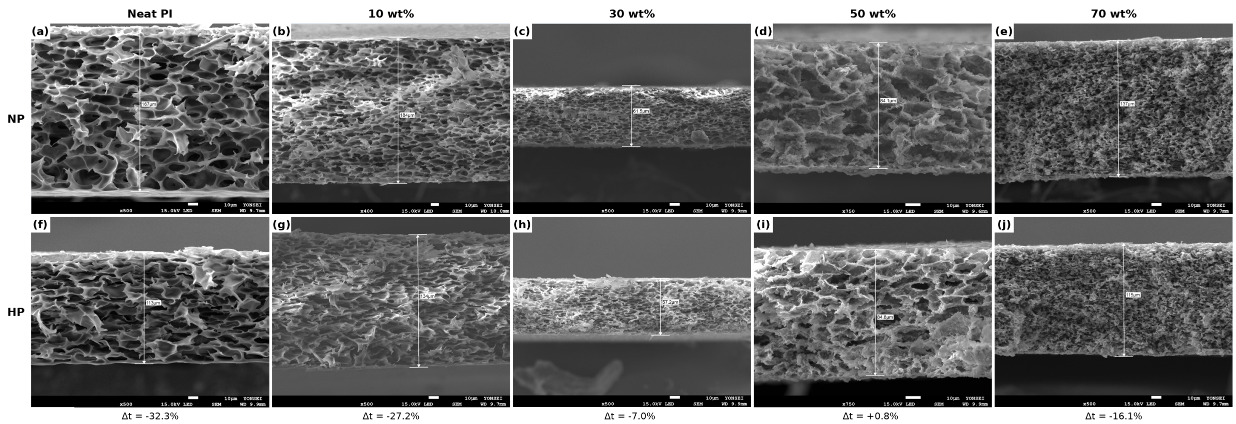

Figure 1: Cross-sectional FE-SEM micrographs of PI/h-BN films at 0–70 wt% h-BN in the non-pressed (a–e, top row) and hot-pressed (f–j, bottom row) states. Δt denotes the film-thickness change upon hot-pressing~~; asterisks mark the 30 and 50 wt% compositions flagged for anomalous porosity behavior (see text)~~. Top-view micrographs and EDS elemental analysis are provided in ~~Supplementary Figures~~Figure ~~S3~~ S7 and Table S1.

~~Hot-Pressed (HP) films are produced by applying uniaxial compressive force at elevated temperature to NP films.~~ Hot-pressing ~~selectively collapses portions~~ collapses part of the open pore network ~~, increasing~~ and raises bulk density and tensile strength. ~~For most compositions , HP porosities are substantially lower than NP~~ At most compositions the HP open porosity is far below the NP value (43.4% vs. 75.4% for neat PI; 29.8% vs. 91.4% ~~for~~ at 10 wt%~~BN). However, the~~), but at 30 ~~wt%~~ and 50 wt% ~~HP films exhibit anomalously higher~~ the HP films show higher intrusion porosities than their NP counterparts (67.2% vs. 48.8%~~and~~ ; 64.6% vs. 58.1%)~~, accompanied by near-zero film thickness reduction at 50 wt% (−0.8%). This behavior is attributed to structural redistribution of h-BN platelets under compressive load: the rigid BN network resists overall compaction while internal pore connectivity changes increase the measurable open-channel porosity. Re-measurement of these two compositions by an independent technique (e.g. , gas~~

~~pycnometry)is recommended to resolve this anomaly.~~ ~~[editorial comment removed]~~ ~~To assess the sensitivity of the LN model to this uncertainty, predictions were compared using the measured anomalous HP porosities versus their expected (lower) values: the resulting change in predicted was ±0.12 at most (for~~ . Mercury intrusion measures only the open, connected pore volume. The total porosity from the bulk density, $\phi_{\text{total}} = 1 - \rho/\rho_{\text{solid}}$ (Table 3), exceeds the intrusion value for six of the seven films with measured ρ, by up to 31 percentage points (NP 30 wt%~~HP), which is within the overall model RMSE and does not qualitatively alter any conclusion. The inter-laboratory dielectric comparison at this same composition (Section 3.4) independently corroborates the anomaly. SEM cross-sections (Figure 1)confirm progressive densification at 0, 10, and~~ : 79.8% vs. 48.8%), and the difference is a measure of closed porosity. Notably, ϕ_{total} falls from NP to HP at every composition (30 wt%: 79.8 $\rightarrow$ 71.6%; 50 wt%: 70.6 $\rightarrow$ 61.3%; 70 wt%: 72.8 $\rightarrow$ 72.1%). The reversed ordering of the intrusion values at 30 and 50 wt% ~~BN (~~therefore reflects a change in pore connectivity, in which compaction opens previously closed pores to intrusion, not an increase in pore volume. Both quantities are listed in Table 3; the choice of porosity input to Eq. 3 is discussed in Section 4.2. The SEM cross-sections agree: thickness falls by 32.3%, 27.2%, and 16.1% ~~thickness reduction, respectively), while the 30 and 50~~at 0, 10, and 70 wt%~~films show near-zero thickness change (~~, but changes by only −7.0% and +0.8% ~~) — the microstructural counterpart of the anomalous porosity behavior. EDS elemental analysis (Supplementary~~ at 30 and 50 wt%, consistent with the resistance of the h-BN network to compaction at these loadings.

EDS (Table S1) ~~confirms that~~ shows the boron content ~~scales systematically~~ rising with nominal loading (~~B: 5.4~~ 8.0 to 29.3 wt% ~~from 0~~ B from 10 to 70 wt% h-BN, NP series)~~and is preserved upon~~ , preserved after hot-pressing, ~~verifying composition retention through the NIPS and densification steps.~~

~~EDS elemental mapping confirms near-quantitative h-BN incorporation across all compositions, with measured boron weight fractions~~ and within ±5 wt% (absolute) of ~~theoretical values~~the theoretical values at ≥10 wt%; the apparent 5.4 wt% B in the neat film is the quantification limit of EDS for light elements and is treated as background. Spot-to-spot ~~variance (<~~variation below 1 wt% B at 70 wt% ~~BN) confirms homogeneous h-BN dispersion achieved by~~ confirms that the sonication-assisted NIPS protocol gives a homogeneous dispersion.

4.2. Dielectric ~~Constant: Experiment and 3-Phase LN Model~~constant

Figure ~~?? presents~~ 2a compares the measured dielectric constants ~~and 3-phase LN model~~ with the three-phase LN predictions for all ~~10 samples. All films exhibit~~ ten films (parity plot in Figure S2). Every film lies far below dense PI (∼3.5)~~, confirming that NIPS-generated porosity dominates the dielectric response. NP films achieve~~ : the NP films span ε_r = 1.28–1.76 ~~(100 kHz); HP films span $\varepsilon_r = 1.62$~~at 100 kHz and the HP films 1.62–2.18~~. The monotonic decrease of ε_{NP} with increasing BN loading (1.76 → 1.28) reflects increasing BN solid content combined with generally high porosity . Hot-pressing uniformly~~ , so NIPS porosity dominates the dielectric response, and hot-pressing raises ε_r by ~~$\Delta\varepsilon_r = 0.18$~~0.18–0.42 at every composition through pore collapse. In the NP series ε_r falls monotonically with h-BN content (1.76 $\rightarrow$ 1.28).

Because ε_{BN} (~6) exceeds ε_{PI} (~3.5) and the open porosity does not rise over this range, no mixing rule reproduces this trend (Section S3.5); the model predictions, evaluated at each film's open porosity, follow the porosity input rather than the composition.

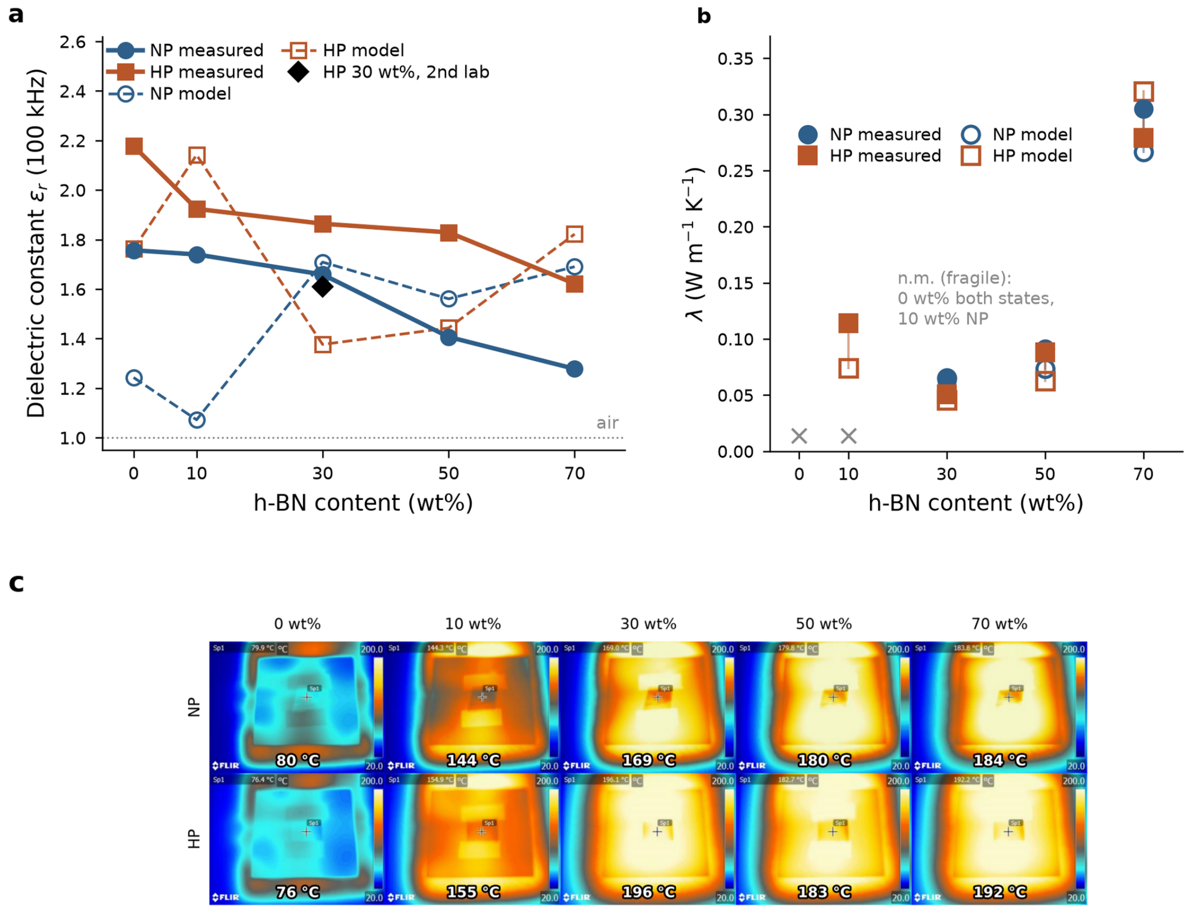

Figure 2: Measured dielectric and thermal properties of PI/h-BN films ($n = 1$ per condition). (a) Dielectric constant ~~(~~at 100 kHz ~~)~~ vs. h-BN content for ~~non-pressed (~~NP ~~)~~ and ~~hot-pressed (~~HP ~~) PI/h-BN composite~~ films~~. Filled~~; filled symbols~~:~~, measured~~values ($n = 1$ specimen per condition). Open~~; open symbols joined by dashed lines~~: 3-phase Lewis Nielsen~~, three-phase LN predictions ~~evaluated~~ at each film's ~~own~~ measured open porosity~~. Black~~; black diamond~~:~~, independent ~~inter-laboratory~~ measurement of the HP 30 wt% film (KOPTRI, ASTM D150), not used in fitting. (b) ~~Parity plot of predicted~~ Thermal conductivity vs. ~~measured dielectric constant for all 10 samples~~h-BN content; open symbols, ~~with ±10% and ±20% bands. The panel title reports the model RMSE alongside the standard deviation of the data itself~~single-parameter LN predictions ($\lambda_{BN,eff}$ fitted; ~~because RMSE exceeds SD~~$A = 5.5$, $\phi_{max} = 0.60$ fixed), joined to the ~~calibrated model does~~ measured value by a tie-line; crosses mark compositions that could not ~~outperform the data mean~~ be measured (~~see Section 4.2~~film fragility). (c) Infrared thermography of all ten films on a 200 °C stage after [X] min equilibration; annotated values are uncorrected radiometric spot temperatures and are comparative, not absolute. Parity plot and porosity-sensitivity curves are given in Figures S2 and S3.

The ~~LN~~ model returns RMSE = 0.43 (NP) and 0.36 (HP)~~. [editorial comment removed]~~, larger than the standard deviation of the measured ε_r across compositions (0.19 NP, 0.18 HP), so the coefficient of determination is negative ($R^2 = -3.9$ NP, -3.0 HP; -1.6 for the 30–70 wt% subset). The calibrated model does not predict ε_r for an individual film better than the data mean, and we use it accordingly: as a mechanistic description of how porosity and pore geometry enter ε_r, and as a generator of design maps whose absolute levels carry an uncertainty of the order of the RMSE. Part of the residual lies in the porosity input. The fits use the open porosity, which is available for all ten films; ε_r and λ respond to the total air fraction regardless of connectivity, so ϕ_{total} is the physically appropriate input, but it exists only for the seven films with measured density and differs from the open porosity by up to 31 percentage points (Section 4.1). The largest residuals ~~occur~~ are at 0~~and~~, 10, and 70 wt% ~~BN (model underprediction of~~ in the NP series: underprediction by 29–38% ~~,~~ at 0–10 wt% and overprediction by 32% at 70 wt% (Table 2). ~~Two physical mechanisms explain this systematic~~

Two mechanisms account for the underprediction at low ~~BN~~ loading. First, ~~at these compositions,~~ the NP films ~~exhibit extremely high porosities (~~at 0 and 10 wt% have porosities of 75.4% and 91.4%~~, respectively). At~~ ~~$\phi_p > 0.75$, the sponge pore network transitions from a regime of moderate connectivity(~~. Above $\phi_p \approx 0.75$ the sponge passes from moderate connectivity, where the mean-field $A_{\text{pore}} = 0.08$ ~~provides adequatedescription) to a near-fully-open framework where~~ is adequate, to a nearly open framework in which the PI skeleton itself becomes discontinuous ~~. In this regime, the effective dielectric constant is dominated by~~ and long-range ~~pore connectivity and wall-thinning effects that are outside the scope of a local mean-field mixingrule~~connectivity and wall thinning, not local mixing, set the response. Second, at 10 wt% ~~BN with~~ and 91.4% porosity ~~, the actual BN~~ the h-BN volume fraction in the film is only $\phi_{\text{BN,film}} = 0.006$~~—effectively negligible—meaning the dielectric response should approach that of pure PI foam. For the limiting inverted topology (~~, so the film should behave as a PI foam; yet the opposite limit, isolated PI inclusions in air~~), the dilute Maxwell–Garnett estimate~~ ~~$\varepsilon_r \approx 1 + 3\phi_s(\varepsilon_{\text{PI}} - 1)/(\varepsilon_{\text{PI}} + 2)$ at solid fraction $\phi_s \approx 0.087$.~~, gives ~~$\varepsilon_r \approx 1.12$ —~~ $\varepsilon_r \approx 1 + 3\phi_s(\varepsilon_{\text{PI}} - 1)/(\varepsilon_{\text{PI}} + 2) \approx 1.12$ at $\phi_s \approx 0.087$, also far below the measured 1.74. The ~~NIPS film at this extreme porosity likely exhibits a partially inverted topology that neither the Maxwell–Garnett nor the Lewis–Nielsen mean-field picture captures: the measured value indicates that the real PI skeleton remains substantially~~ real skeleton is evidently more connected than either idealized topology~~assumes~~, a partially inverted structure that no mean-field picture captures. These limitations are ~~inherent to mean-field approaches~~ confined to the high-porosity end and do not affect ~~conclusions at compositions with moderate porosity(~~the conclusions at moderate porosity: for the NP films of 30–70 wt% ~~BN range), where restricting the RMSE evaluation to 30–70~~the RMSE is 0.25.

An independent measurement of the HP 30 wt% ~~gives 0.25 (NP)~~film at KOPTRI (ASTM D150) gives $\varepsilon_r = 1.610$, against 1.864 in-house (−13.6%) and a model value of 1.376 at the open porosity of 67.2% (Figure 2a, black diamond; Table 2); the three values disagree pairwise at the 14% level. The inter-laboratory difference estimates the reproducibility of ε_r for a porous film with contact electrodes: an air gap between a rough surface and the electrode acts as a series capacitance that lowers the apparent ε_r by an amount that depends on surface texture, and therefore on composition and processing state. This effect is a plausible experimental contributor to the negative R^2 and to the composition trend of the NP series, and re-measurement with sputtered or painted electrodes would settle it. The model residual at 30 wt% HP (−26%), however, exceeds the inter-laboratory spread and is a model limitation rather than a measurement artefact.

Table 2: Experimental vs. predicted dielectric constants (~~3-phase~~ three-phase Lewis–Nielsen model). KOPTRI: independent inter-laboratory measurement of the HP 30 wt% film (ASTM D150; not used in fitting).

h-BN (wt%)	~~ε_{NP}~~ $\varepsilon_{r,NP}$ (Exp.)	~~ε_{NP}~~ $\varepsilon_{r,NP}$ (LN)	Δ (%)	~~ε_{HP}~~ $\varepsilon_{r,HP}$ (Exp.)	~~ε_{HP}~~ $\varepsilon_{r,HP}$ (LN)	Δ (%)	KOPTRI (HP)
0	1.757	1.242	−29.3	2.178	1.760	−19.2	—
10	1.740	1.073	−38.4	1.924	2.139	+11.2	—
30	1.659	1.707	+2.9	1.864	1.376	−26.2	1.610*
50	1.407	1.560	+10.8	1.829	1.442	−21.2	—
70	1.278	1.689	+32.1	1.622	1.819	+12.2	—

~~$\Delta = (\varepsilon_{LN} - \varepsilon_{Exp})/\varepsilon_{Exp} \times 100\%$~~ $\Delta = (\varepsilon_{r,LN} - \varepsilon_{r,Exp})/\varepsilon_{r,Exp} \times 100\%$. LN predictions are evaluated at each film's open porosity. *Inter-laboratory deviation from the in-house HP value (1.864) is −13.6%; the model prediction at the ~~measured (anomalous)~~ open porosity of 67.2% is 1.376 ~~. All three values converge if the porosity lies in the expected 40–45% range — see~~ (Section ~~3.4~~4.2).

4.3. Thermal ~~Conductivity: Experiment and Lewis–Nielsen Fit~~conductivity

Figure ~~??a shows thermal conductivity results~~2b shows the thermal conductivities, and Table 3 ~~lists the underlying laser-flash data (bulk~~ the underlying density, specific heat, and ~~thermal diffusivity). Measurements were unavailable for the neat films in both states and for~~ diffusivity together with the open and total porosities. The neat films and the NP 10 wt% ~~due to film fragility at these high-porosity compositions (film~~ were too fragile to measure (porosities up to 91.4%), leaving seven ~~measured~~ data points. ~~For all available data, the~~ The single-parameter LN calibration reproduces ~~the trends well~~ all of them (RMSE$_{NP}$ = 0.027, RMSE$_{HP}$ = 0.030 W m^{-1} K^{-1}): ~~near-constant~~ λ ~~at 30–50~~is nearly constant from 30 to 50 wt% ~~BN and a sharp increase~~ and rises sharply at 70 wt%. ~~Within the model , the 70 wt% upturn arises naturally from the packing-interaction~~ In the model the rise comes from the packing term ψ in Eq. 1~~:~~ , since at $\phi_{BN} = 0.591 \approx 0.99\,\phi_{max}$ ~~, the composite approaches the maximum platelet packing~~ the composite is at the platelet packing limit, which is the ~~Lewis–Nielsen~~ LN representation of imminent particle–particle contact~~(percolation)~~.

~~(a) Thermal conductivity vs. h-BN content: measured laser-flash values (filled symbols, $n = 1$ per condition) and 3-phase LN predictions evaluated at each film's own measured porosity (open symbols), joined by a vertical tie-line so the per-sample deviation is directly readable; the single-parameter thermal calibration is used throughout (fitted; $A = 5.5$, $\phi_{max} = 0.60$ fixed). Crosses mark compositions for which could not be measured (film fragility): neat PI in both states and NP 10 wt%. (b) Porosity sensitivity at fixed 50 wt% h-BN: (left axis, solid) and (right axis, dashed) both rise as porosity falls, so porosity alone cannot decouple the two properties; circles and squares mark the measured NP and HP 50 wt% films. (c) Infrared thermography of all ten films on a 200 °C stage, labeled by composition (columns) and processing state (rows); the annotated value is the Sp1 spot temperature. [editorial comment removed]~~

The fitted $\lambda_{BN,eff}$ = 6.81 W m^{-1} K^{-1} ~~is approximately 7× lower than the bulk~~ lies well below the through-plane ~~h-BN conductivity~~ conductivity of bulk h-BN (~~~50 W m^{-1} K^{-1}[editorial comment removed]), attributable primarily to Kapitza thermal interface resistance (TIR)~~ ∼[value] W m^{-1} K^{-1} [14]). We attribute the reduction mainly to Kapitza resistance at the h-BN–PI interface~~. At 70 nm particle diameter, the~~

Table 3: ~~Thermophysical~~ Porosity and thermophysical data underlying the thermal-conductivity determination ($\lambda = \alpha\rho C_p$; laser flash at 25 °C). ϕ_{open}: mercury intrusion; $\phi_{\mathrm{total}} = 1 - \rho/\rho_{\mathrm{solid}}$ with $\rho_{\mathrm{PI}} = 1.42$ and $\rho_{\mathrm{BN}} = 2.29\,\mathrm{g\,cm^{-3}}$.

h-BN (wt%)	State ~~State~~	ϕ_{open} ~~ρ~~ (%)	ϕ_{total} (%)	ρ ($\mathrm{g\,cm^{-3}}$)	C_p ~~C_p~~ ($\mathrm{J\,g^{-1}\,K^{-1}}$)	α ~~α~~ ($\mathrm{mm^2\,s^{-1}}$)	λ ($\mathrm{W\,m^{-1}\,K^{-1}}$)
0	NP	75.4	—	—	—	—	—
0	HP	43.4	—	—	—	—	—
10	NP	91.4	—	—	—	—	—
10	HP	29.8	40.4	0.879	0.903	0.143	0.114
30	NP	48.8	79.8	0.323	0.853	0.230	0.063
30	HP	67.2	71.6	0.455	0.863	0.131	0.051
50	NP	58.1	70.6	0.515	0.875	0.202	0.091
50	HP	64.6	61.3	0.678	0.885	0.147	0.088
70	NP	54.2	72.8	0.526	0.838	0.693	0.305
70	HP	48.8	72.1	0.539	0.846	0.613	0.279

Neat PI (both states) and NP 10 wt% could not be measured by laser flash (film fragility), and ρ was not determined for these films. The LN fits use ϕ_{open} as the porosity input (Section 4.2).

~~surface-area-to-volume ratio is approximately 30 m^2 g^{-1}, causing~~ : at 70 nm the surface-to-volume ratio is about 30 m^2 g^{-1}, so interfacial phonon scattering ~~to limit~~ limits heat transfer through ~~individual particles — consistent in order of magnitude with molecular-dynamics-based interfacial conductance estimates for BN–polymer~~ each particle. Molecular-dynamics estimates of the interfacial conductance of h-BN–polymer interfaces ($G \sim 10$–~~100 MW/m^2/K) , which at 70 nm correspond to~~ 100 MW m^{-2} K^{-1}) translate at 70 nm into effective particle conductivities of ~~order 1–7 W/m K [30]. Two residual features~~ 1–7 W m^{-1} K^{-1} [30], consistent in magnitude with the fitted value. Two residuals carry physical information, with the caveat that both are comparable to the ∼10% propagated uncertainty in λ (Section 2.5). First, the NP 70 wt% film exceeds the ~~model~~ prediction (0.305 vs. 0.263 W m^{-1} K^{-1}, +16%), consistent with incipient ~~BN–BN contact percolation~~ h-BN–h-BN contact beyond the mean-field description. Second~~ — and experimentally striking —~~, hot-pressing at 70 wt% ~~reduces~~ lowers λ ~~(0.305 → 0.279 W m^{-1} K^{-1}) despite lowering porosity (54.2% → 48.8%), opposite to the model trend (which predicts an increase~~ from 0.305 to 0.279 W m^{-1} K^{-1} although the open porosity falls (54.2 → 48.8%) and the total porosity is nearly unchanged (72.8 → 72.1%); the model predicts a rise to 0.315~~): this indicates that compaction partially disrupts the BN–BN~~. Compaction evidently disrupts part of the h-BN contact network formed during NIPS, sacrificing percolative pathways even as the solid fraction increases. ~~Porosity sensitivity at fixed composition~~ At fixed composition the two properties remain coupled through porosity (Figure ~~??b)shows that both properties remain strongly porosity-coupled~~ S3): at 50 wt%~~: over the experimentally~~, over the accessible 30–65% porosity range, ε_r spans 1.43–2.31 ~~while~~ and λ ~~spans~~ 0.06–0.16 W m^{-1} K^{-1}.

4.4. ~~Design Maps for Simultaneous Property Optimization~~ Tensile strength and coefficient of thermal expansion

Hot-pressing raises the tensile strength by 19–95% at every composition (Figure 3a; Table 4), as expected for pore collapse under compressive load. The NP strength is non-monotonic, with a maximum at 30 wt% (25.4 MPa) and minima at 10 wt% (14.9 MPa) and 50 wt% (12.5 MPa), and follows the porosities at these compositions (91.4% and 58.1%) inversely. The LN + Gibson–Ashby model reproduces this trend qualitatively.

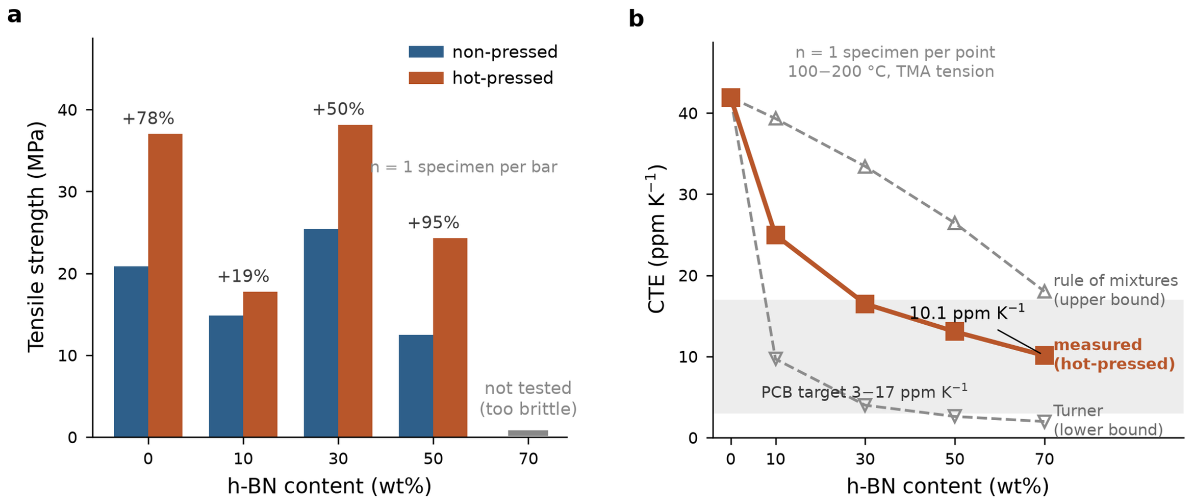

Figure 3: Mechanical and thermal-expansion behavior of PI/h-BN films. (a) Tensile strength vs. h-BN content for NP and HP films ($n = 1$ specimen per bar). Percentage values indicate the NP→HP change at each composition; the 70 wt% slot carries a baseline stub because those films were too brittle to test, not because the strength is zero. (b) CTE vs. h-BN content for HP films ($n = 1$ per point; TMA, tension mode, 100–200 °C): measured values against the Turner lower bound and the rule-of-mixtures (ROM) upper bound, both computed with $\alpha_{PI} = 41.92\,\mathrm{ppm\,K^{-1}}$ (measured). The PCB substrate target zone (3–17 $\mathrm{ppm\,K^{-1}}$) is shaded. The measured CTE lies between the two bounds because the discontinuous load-path geometry of the porous sponge gives only partial elastic constraint (Section 3.3).

The measured CTE falls from 24.96 to 10.11 $\mathrm{ppm\,K^{-1}}$ between 10 and 70 wt% (Figure 3b). The Turner model, with $\alpha_{PI} = 41.92\,\mathrm{ppm\,K^{-1}}$ from the neat PI HP film, reproduces the neat value but underpredicts by 61–80% at every loading, and the ROM overpredicts because it ignores elastic constraint. The data lie between the two bounds, as expected for partial constraint in a sponge whose platelets sit in discontinuous PI ligaments rather than in a continuous dense matrix. Dense-composite CTE models therefore reach their limit for sponge materials, and an open-cell foam mechanics treatment is a natural next step. At 70 wt% (HP) the CTE of 10.1 $\mathrm{ppm\,K^{-1}}$ lies within the PCB substrate window of 3–17 $\mathrm{ppm\,K^{-1}}$. **[CTE was measured on the HP films o**

Table 4: Tensile strength and CTE data for PI/h-BN composite films.

h-BN (wt%)	NP strength (MPa)	HP strength (MPa)	NP→HP (%)	HP CTE ($\mathrm{ppm\,K^{-1}}$)	Turner ($\mathrm{ppm\,K^{-1}}$)	ROM ($\mathrm{ppm\,K^{-1}}$)
0	20.85	37.02	+77.6	41.92	41.92	41.92
10	14.86	17.74	+19.4	24.96	9.69	39.31
30	25.43	38.15	+50.0	16.49	4.00	33.43
50	12.47	24.28	+94.7	13.06	2.61	26.45
70	—*	—*	—	10.11	1.98	18.02

*Not tested (film brittleness at 70 wt% h-BN). Turner and ROM use $\alpha_{PI} = 41.92\,\mathrm{ppm\,K^{-1}}$ (measured neat porous PI, HP), $E_{PI} = 3.5\,\mathrm{GPa}$, $E_{BN} = 200\,\mathrm{GPa}$, $\alpha_{BN} = 1.5\,\mathrm{ppm\,K^{-1}}$, solid-phase volume fractions.

4.5. Design maps for simultaneous property optimization

Figure ~~?? presents model-generated design maps of~~ 4a,b maps the model ε_r and λ ~~spanning the full composition space (h-BN content~~ over the full design space (0–70 wt% ~~; porosity~~ h-BN, 0–95% porosity), with ~~experimental~~ the measured NP and HP states overlaid and arrows ~~indicating~~ marking the NP→HP ~~processing~~ trajectory at each composition.

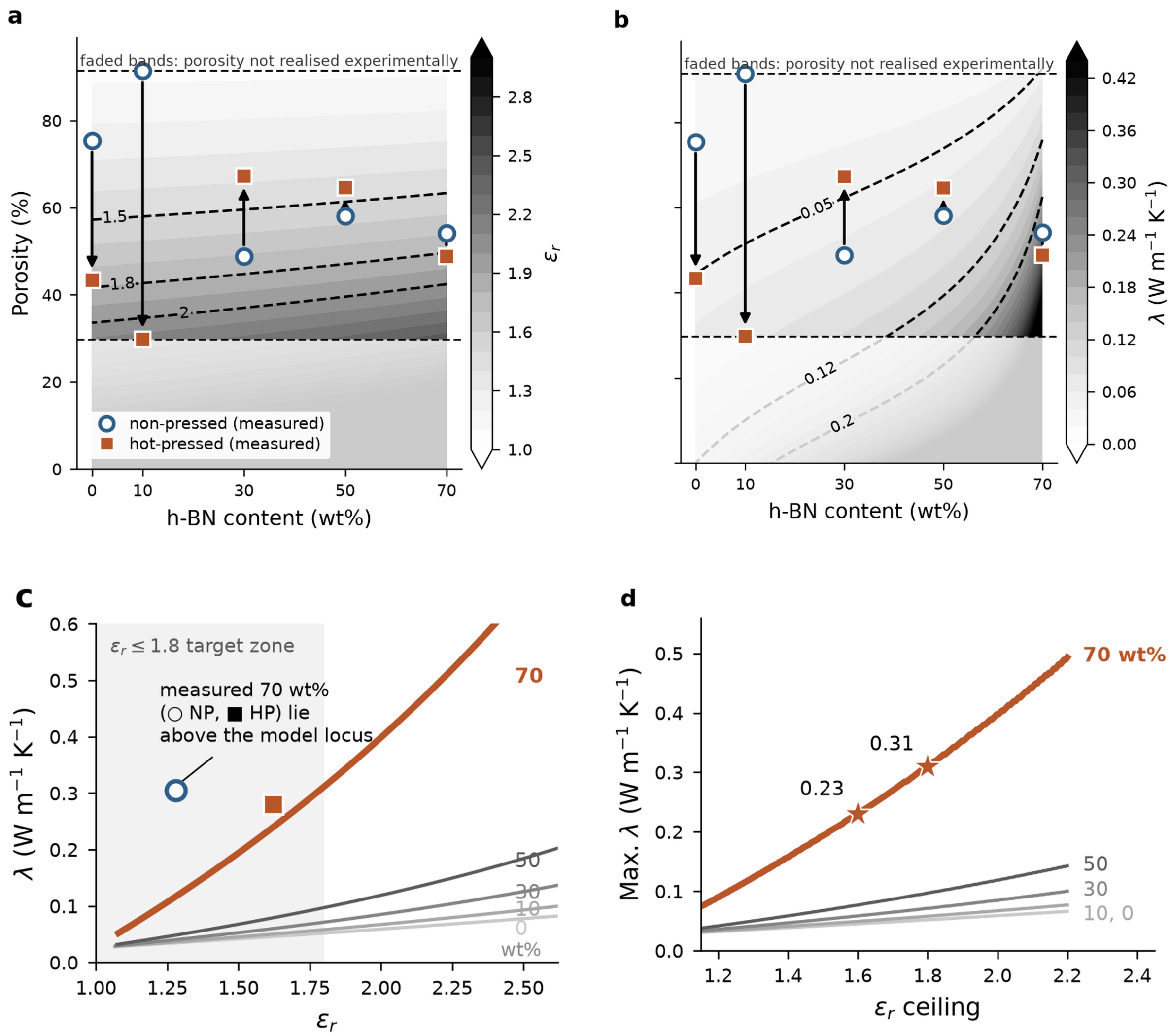

Figure 4: ~~Design maps~~ Model-guided design (~~3-phase~~ three-phase LN model, calibration of Table 1)~~for~~. (a, b) ~~dielectric constant~~ Design maps of ε_r and ~~(b) thermal conductivity~~ λ ~~(W/m K) as functions of~~ vs. h-BN content and porosity~~. Measured~~; measured NP (open circles) and HP (filled squares) states ~~are~~ overlaid~~;~~, arrows ~~indicate~~ indicating the NP→HP ~~processing~~ trajectory~~. The~~; faded bands above ~~91 %~~ 91% and below ~~30 %~~ 30% porosity lie outside the range ~~realised~~ realized by any ~~of~~ film. (c) ε_r–λ loci for each loading with porosity swept along each curve; the ~~ten~~ measured 70 wt% films ~~, so~~ lie above their own locus. (d) Maximum λ attainable under a given ε_r ceiling with porosity optimized at each point; stars mark the ~~map there is extrapolation rather than interpolation between measured states~~ $\varepsilon_r \le 1.6$ and $\varepsilon_r \le 1.8$ operating points at 70 wt%. The absolute levels of panels (a)–(d) inherit the dielectric uncertainty quantified in Section 4.2. Feasible-window and porosity-optimum panels are given in Figures S4 and S5.

Three ~~actionable design principles emerge~~design rules follow. First, ~~achieving~~ $\varepsilon_r < 1.5$ requires a porosity above ∼57% at low ~~BN loading , rising to~~ loading and ∼63% at 70 wt%~~ — comfortably accessible to NIPS~~

~~in both~~, a range that NIPS reaches in both the NP and moderately pressed states. Second, λ is ~~strongly loading-gated~~gated by loading: $\lambda > 0.20\,\mathrm{W\,m^{-1}\,K^{-1}}$ is ~~unreachable~~ out of reach below ~55 wt% ~~BN~~ at any practical porosity, ~~requires~~ needs a porosity below ~35% at 60 wt%, but is available up to ~62% porosity at 70 wt%~~—~~, because the packing-limit amplification at high loading ~~dramatically~~ relaxes the porosity constraint. Third, ~~the NP→HP trajectories show that~~ hot-pressing ~~traverses the maps nearly vertically (porosity is the processing variable)~~moves a film almost vertically on the maps, so composition must be ~~selected~~ chosen first and porosity tuned second; at 70 wt% ~~,~~both the NP and HP ~~states~~ films already lie in the ~~simultaneously favorable region~~ region favorable for both properties.

4.6. ~~*Shape Factor Sensitivity Analysis*~~*Shape-factor sensitivity*

~~Supplementary~~ Figure S1 ~~systematically~~ examines the sensitivity of ~~LN~~ the predictions to the h-BN shape factor ~~A for values 0.3~~over $A = 0.3$–6.0, ~~corresponding to geometries~~ from needle-like rods to ~~highly~~ aligned platelets.

For the dielectric model (Figure S1a) ~~, all A~~ the curves are nearly indistinguishable~~—confirming that~~: ε_r is ~~dominated by the Step 2~~ set by the pore term (step 2, Eq. 3), not by A_{BN}~~. In~~, so in porous NIPS films ~~, BN particle geometry is essentially irrelevant to dielectric performance; only porosity level~~ the particle geometry has little influence and only the porosity and pore architecture (A_{pore}) matter. This justifies treating the fitted dielectric A_{BN} as weakly determined (Table 1) and confirms that the ~~RMSE values~~ residuals in Table 2 reflect mean-field ~~model~~ limitations rather than parameter uncertainty.

~~For the~~ The thermal model (Figure S1b) ~~, sensitivity~~ is more sensitive to A ~~is higher~~ above 30 wt%~~BN~~, which is ~~precisely why~~ why we fixed A ~~was fixed~~ at its literature platelet value rather than ~~fitted~~ fitting it (Section 3.4). At the NP mean porosity ~~(~~of 65.6%~~), all~~, every curve from $A = 0.3$ ~~–6.0 curves remain below the experimental~~ to 6.0 stays below the measured NP 70 wt% value (0.16–0.18 vs. $0.305\,\mathrm{W\,m^{-1}\,K^{-1}}$), ~~reinforcing that~~ so the 70 wt% enhancement ~~is driven by~~ comes from the proximity to maximum packing and incipient particle contact, not ~~by any plausible choice of~~ from any single-particle shape factor. ~~In the practically important~~ Over 30–50 wt% ~~range, the A~~ the curves converge within $\pm 0.02\,\mathrm{W\,m^{-1}\,K^{-1}}$, and the mean-field ~~LN~~ model is adequate.

4.7. ~~*Mechanical Properties*~~ *Property trade-offs* and ~~*Coefficient of Thermal Expansion*~~*design guidance*

~~Figure ?? presents tensile strength and CTE results. Hot-pressing consistently improves tensile strength by 19–95% (Table 4), consistent with pore collapse under compressive load. The non-monotonic NP tensile strength—local maximum at 30 wt% (25.4 MPa), minima at 10 wt% (14.9 MPa) and 50 wt% (12.5 MPa)—correlates inversely with the anomalous porosity at 10 wt% (91.4%) and high porosity at 50 wt% (58.1%), respectively. The LN + Gibson–Ashby model reproduces this trend qualitatively.~~

~~(a) Tensile strength vs. h-BN content for NP and HP films ($n = 1$ specimen per bar). Percentage values indicate the NP→HP change at each composition; the 70 wt% slot carries a baseline stub because those films were too brittle to test, not because the strength is zero. (b) CTE vs. h-BN content for HP~~

~~films ($n = 1$ per point; TMA, tension mode, 100–200 °C): measured values against the Turner lower bound and the rule-of-mixtures (ROM) upper bound, both computed with $\alpha_{\mathrm{PI}} = 41.92$ ppm/K (measured). PCB substrate target zone (3–17 ppm/K) is shaded. The measured CTE lies between the two bounds because the discontinuous load-path geometry of the porous sponge gives only partial elastic constraint (Section 3.3).~~

~~For CTE (Figure ??b), the Turner model (using $\alpha_{\mathrm{PI}} = 41.92$ ppm/K as measured from the neat PI HP sample) correctly predicts the neat-film value but substantially underpredicts CTE at all BN loadings (−61 to −80%). The ROM overestimates because it neglects elastic constraint entirely. The experimental CTE values (24.96 → 10.11 ppm/K from 10 → 70 wt%) lie between the two model bounds, physically consistent with partial elastic constraint in a porous sponge where h-BN platelets are embedded within discontinuous PI ligaments rather than a macroscopically continuous dense matrix. This porous-architecture effect is an important limitation of applying dense-composite CTE models to sponge materials and represents an opportunity for future model development incorporating open-cell foam mechanics. Critically, the measured CTE = 10.1 ppm/K at 70 wt% h-BN (HP) lies within the PCB substrate compatibility window (3–17 ppm/K), confirming viability for copper-metallized flexible packaging applications.~~

~~Tensile strength and CTE data for PI/h-BN composite films. h-BN NP strength HP strength NP→HP HP CTE Turner ROM (wt%) (MPa) (MPa) (%) (ppm/K) (ppm/K) (ppm/K) 0 20.85 37.02 +77.6 41.92 41.92 41.92 10 14.86 17.74 +19.4 24.96 9.69 39.31 30 25.43 38.15 +50.0 16.49 4.00 33.43 50 12.47 24.28 +94.7 13.06 2.61 26.45 70 — * — * 10.11 1.98 18.02~~ ~~*Not tested (film brittleness at 70 wt% h-BN). Turner and ROM use $\alpha_{\mathrm{PI}} = 41.9$~~ ~~neat porous PI, HP), $E_{\mathrm{PI}} = 3.5$ GPa, $E_{\mathrm{BN}} = 200$ GPa, $\alpha_{\mathrm{BN}} = 1.5$ ppm/K, solid-pha~~

4.8. ~~*Property Trade-offs and Composite Design Guidance*~~

For each ~~h-BN loading , the calibrated LN~~ loading the calibrated model traces a ~~continuous~~ locus in ε_r–λ space as porosity is swept (Figure ~~??b~~4c); the 70 wt% HP ~~state already lies within the target zone~~ film ($\varepsilon_r = 1.62$, $\lambda = 0.28\,\mathrm{W\,m^{-1}\,K^{-1}}$) ~~.~~

~~Three design insights emerge. First, along any fixed-composition locus, reducing porosity raises both~~ already lies in the target zone. Three points follow. Along any locus, lowering porosity raises ε_r and λ together~~—the two properties are fundamentally~~ ; the two are coupled through the same structural variable~~and cannot be independently tuned by porosity control alone; every porosity choice~~ , and every porosity is a position on a trade-off curve. ~~Second, increasing BN~~ Raising the loading shifts the ~~entire locus toward~~ whole locus to higher λ at ~~any~~ a given ε_r, ~~making composition the primary~~ so composition is the lever for thermal performance at ~~constant~~ a fixed dielectric specification. ~~Third, the~~ The 70 wt% locus ~~passes~~ runs through the most favorable region: ~~experimentally,~~ the NP film at 54.2% open porosity ($\varepsilon_r = 1.28$, $\lambda = 0.31\,\mathrm{W\,m^{-1}\,K^{-1}}$) meets all three target levels defined below, and the HP film at 48.8% ~~porosity~~ ($\varepsilon_r = 1.62$, $\lambda = 0.28\,\mathrm{W\,m^{-1}\,K^{-1}}$) ~~bracket an operating window in which all three target levelsdefined below are met or exceeded~~meets the Basic and Advanced levels.

4.8. ~~*Model-Guided Computational Design*~~*Model-guided design*: ~~*Performance Limits*~~ *performance limits and* ~~*Processing Prescriptions*~~*processing prescriptions*

~~A key advantage of a calibrated physics-based model over purely empirical data is the ability to perform systematic computational optimization — exploring~~ The calibrated model can be interrogated over the full (~~h-BN~~ loading, porosity) ~~design space to identify theoretical~~ space for performance limits and for the processing conditions ~~required to reach any specification. This constitutes model-guided composite design: rather than requiring exhaustive trial-and-error experimentation, the calibrated 3-phase LN model provides a predictive surface that can be interrogated algorithmically across thousands of composition–processing combinations.~~

~~Figure ?? presents a three-part analysis. Panel (a)identifies the feasible processing window — the porosity range simultaneously satisfying three progressively demanding specifications~~ that reach a given specification (Figure 4c,d); the absolute levels of these surfaces inherit the dielectric uncertainty of Section 4.2.

We define three specifications of increasing difficulty: Basic ($\varepsilon_r < 2.0$, $\lambda > 0.05\,\mathrm{W\,m^{-1}\,K^{-1}}$), Advanced ($\varepsilon_r < 1.8$, $\lambda > 0.08\,\mathrm{W\,m^{-1}\,K^{-1}}$), and Future ($\varepsilon_r < 1.6$, $\lambda > 0.12\,\mathrm{W\,m^{-1}\,K^{-1}}$). The porosity window that satisfies each is shown in Figure S4. The Basic window is open ~~across the entire composition range (even~~ at every composition (neat porous PI qualifies at ∼34–44% porosity, since $\lambda_{\mathrm{PI}} = 0.12\,\mathrm{W\,m^{-1}\,K^{-1}}$ ~~exceeds the Basic~~ clears the threshold at moderate porosity); the Advanced window opens at ≥40 wt%~~BN~~; and the Future window opens only at ≥64 wt%~~BN~~, spanning 58–76% porosity at 70 wt%. Experimentally, the 70 wt% NP film ($\varepsilon_r = 1.28$, $\lambda = 0.31\,\mathrm{W\,m^{-1}\,K^{-1}}$) ~~already satisfies~~ meets the Future specification ~~outright, and the 70 wt% HP film satisfies the Advanced specification; the~~ and the HP film meets the Advanced one. The model is conservative ~~for~~ in ε_r at this composition (~~predicting~~ 1.69 predicted vs. ~~the measured~~ 1.28 measured), so the ~~experimental~~ films outperform the model envelope ~~— the prescribed windows should therefore~~ and the windows should be read as sufficient, not necessary, conditions. ~~Panel (b) shows the trade-off loci for all loadings together with the global Pareto frontier of the material system,~~

Figure 4c overlays the loci of all loadings with the Pareto frontier computed over the full ~~(loading, porosity) grid:~~ grid; the frontier coincides with the 70 wt% locus ~~over essentially the entire accessible range, indicating~~ across the accessible range. Because the dielectric residuals exceed the spread of the data and the measured 70 wt% films depart from their locus ($\varepsilon_r = 1.28$ vs. 1.69 predicted, NP), we read this as model-level consistency with the experimental finding that 70 wt% ~~h-BN is the Pareto-optimal composition within the calibrated model. [editorial comment removed] Stars mark the recommended operating point on each locus (maximum subject to $\varepsilon_r \leq 1.8$). Panel (c) answers the engineering question 'what is~~ is the most favorable composition tested, not as a validated frontier of the material system beyond the measured range. Figure 4d answers the practical question of the maximum λ ~~achievable~~ at a given ε_r ceiling~~?'~~: at 70 wt% ~~, the model envelope reaches $\lambda = 0.23\,\mathrm{W\,m^{-1}\,K^{-1}}$~~ the envelope reaches $0.23\,\mathrm{W\,m^{-1}\,K^{-1}}$ at $\varepsilon_r \leq 1.6$ (~~porosity~~ 58% porosity), $0.31\,\mathrm{W\,m^{-1}\,K^{-1}}$ at $\varepsilon_r \leq 1.8$ (~~porosity~~ 50%), and $0.39\,\mathrm{W\,m^{-1}\,K^{-1}}$ at $\varepsilon_r \leq 2.0$ (~~porosity~~ 42%). The ~~experimental~~ measured 70 wt% ~~points (NP:~~ films (0.305 ~~; HP: $0.279\,\mathrm{W\,m^{-1}\,K^{-1}}$~~ NP, $0.279\,\mathrm{W\,m^{-1}\,K^{-1}}$ HP) lie on or slightly above the $\varepsilon_r \leq 1.8$ envelope, ~~validating the optimization~~ consistent with the surface in the ~~experimentally probed region~~ region probed.

~~Model-guided design analysis. (a) Feasible processing window: the porosity range meeting each target specification vs. h-BN content (nested shading, light to dark for Basic →Future); measured NP and HP states overlaid. (b) trade-off loci for each h-BN loading, porosity swept along each curve; the 70 wt% locus (highlighted) lies above every other composition at every dielectric ceiling. The measured 70 wt% films sit above their own model locus, so the model is conservative here. (c) Maximum attainable under a given ceiling, porosity optimised at each point; stars mark the $\varepsilon_r \le 1.6$ and $\varepsilon_r \le 1.8$ operating points at 70 wt%. (d) The porosity that maximises under each ceiling: the five composition curves stay within 10 % (absolute porosity) of one another, showing that the porosity target is set almost entirely by the dielectric ceiling while the achievable at that porosity is set by the loading. Panels (b)–(d) are model surfaces conditioned on the calibration of Table 1; see Section 4.2 for the residual analysis that bounds their reliability.~~

~~The optimization surface condenses into simple engineering~~ The surface reduces to simple prescriptions. The porosity ~~required to maximize~~ that maximizes λ under a ~~fixed~~ given ε_r ceiling is ~~remarkably flat —~~ nearly independent of composition (42–50% for $\varepsilon_r \le 1.8$~~and~~, 52–58% for $\varepsilon_r \le 1.6$~~across the entire composition range — meaning the porosity target is set almost entirely by~~; Figure S5), so the dielectric ceiling ~~, while the achievable~~ sets the porosity target and the loading sets the λ ~~at that porosity is set by the BN loading . In terms of specific specifications, achieving~~ obtainable there. In particular, $\varepsilon_r < 1.7$ with ~~$\lambda > 100$ mW/m/K requires~~ $\lambda > 100\,\mathrm{mW\,m^{-1}\,K^{-1}}$ needs ≥60 wt% ~~BN (porosity~~ at 53–54% ~~)~~porosity; $\varepsilon_r < 1.6$ with ~~$\lambda > 120$ mW/m/K requires~~ $\lambda > 120\,\mathrm{mW\,m^{-1}\,K^{-1}}$ needs 70 wt% at 58%~~porosity; and the most demanding specification (~~; and $\varepsilon_r < 1.5$ ~~, $\lambda > 150$ mW/m/K)~~ with $\lambda > 150\,\mathrm{mW\,m^{-1}\,K^{-1}}$ is feasible only at 70 wt% and 63%~~porosity~~. Since the as-fabricated 70 wt% NP film ~~already~~ sits at 54.2%~~porosity, these prescriptions are reachable~~, these targets are reached by mild or partial hot-pressing~~— full~~. Full densification is neither required nor desirable~~, as it would raise~~: it would push ε_r ~~beyond the ceilings (and, per~~ past the ceilings and, as Section 4.3 ~~, risks disrupting the percolative BN network)~~shows, risks breaking the h-BN contact network.

5. Conclusions

~~Porous~~ We fabricated porous PI/h-BN ~~composite films fabricated by NIPS and characterized in both~~ films by NIPS over 0–70 wt% h-BN, characterized them in the non-pressed ~~(NP)~~ and hot-pressed ~~(HP) stateswere analyzed using a calibrated 3-phase~~ states, and described the full dataset with a calibrated three-phase Lewis–Nielsen framework. ~~The following principal conclusions are drawn: NIPS-generated~~ NIPS porosity is the dominant determinant of the dielectric constant: all films ~~achieve~~ lie at $\varepsilon_r = 1.28$–2.18 ~~at 100 kHz~~(100 kHz), far below dense PI (∼3.5), ~~with~~ and hot-pressing ~~raising~~ raises ε_r by 0.18–0.42 through pore collapse. The fitted pore shape factor $A_{\mathrm{pore}} = 0.08$ quantifies the ~~geometric inefficiency of NIPS bicontinuous sponge pores compared~~ low field-shielding efficiency of the bicontinuous NIPS sponge relative to isolated spherical pores ($A = 1.5$) ~~, providing~~ and provides a model-based ~~microstructure~~ fingerprint that distinguishes NIPS from ~~Pickering emulsion-templated~~ Pickering-emulsion-templated porous PI/~~BN composites. A deliberately~~

~~parsimonious~~ h-BN films. Because the dielectric residuals exceed the spread of the data, the model serves as a mechanistic and design tool rather than as a quantitative predictor of ε_r for individual films.

A single-parameter thermal calibration ($\lambda_{\mathrm{BN,eff}} = 6.81\,\mathrm{W\,m^{-1}\,K^{-1}}$, with platelet shape factor and packing fraction fixed at literature values~~; 7:1 data-to-parameter ratio~~) reproduces the ~~thermal data with RMSE $\leq 0.03\,\mathrm{W\,m^{-1}\,K^{-1}}$. The $\approx 7\times$~~ conductivity data with RMSE $\lesssim 0.03\,\mathrm{W\,m^{-1}\,K^{-1}}$. The reduction of $\lambda_{\mathrm{BN,eff}}$ well below bulk h-BN quantifies ~~Kapitza interface~~ the interfacial (Kapitza) resistance at 70 nm particle size, and the sharp rise in λ ~~upturn~~ at 70 wt% emerges ~~naturally~~ from the approach to maximum platelet packing ($\phi_{\mathrm{BN}} = 0.59 \approx 0.99\,\phi_{\mathrm{max}}$) ~~.~~ without an additional percolation parameter. Hot-pressing at 70 wt% ~~reduces~~ lowers λ ~~($0.305 \to 0.279\,\mathrm{W\,m^{-1}\,K^{-1}}$) despite reducing porosity —~~ from 0.305 to 0.279 $\mathrm{W\,m^{-1}\,K^{-1}}$, opposite to the model trend~~— providing direct experimental evidence~~, indicating that compaction partially disrupts the ~~percolative BN–BN~~ h-BN contact network formed during NIPS. ~~Turner and ROM CTE models respectively underpredict and overpredict at ≥10 wt% BN because the porous~~ The measured CTE lies between the Turner and rule-of-mixtures bounds because the sponge breaks the continuous load-path assumption ~~; the experimental CTE lies between these bounds, reaching 10.1 ppm/K~~ of dense-composite models, and reaches 10.1 $\mathrm{ppm\,K^{-1}}$ at 70 wt% ~~—~~ (HP), within the PCB substrate ~~compatibility window(3–17 ppm/K). Global Pareto analysis shows that the~~ window.

Within the compositions tested, 70 wt% h-BN ~~locus constitutes the~~ gives the most favorable ε_r–λ ~~Pareto frontier of the entire material system. The~~ combination, and the calibrated model is consistent with this result: the 70 wt% ~~NP film already satisfies the most demanding target examined ($\varepsilon_r < 1.6$, $\lambda > 0.12\,\mathrm{W\,m^{-1}\,K^{-1}}$) experimentally~~locus lies above every other composition across the accessible porosity range, and the model prescribes ~~the porosity window (~~a porosity window of 50–63% at 70 wt%, reachable ~~by mild hot-pressing~~ from the as-fabricated 54% ~~) required to navigate the trade-off toward any intermediate specification.~~ by mild hot-pressing. These results establish the pore shape factor as a microstructural design parameter for porous ceramic-filled polymer films and provide a calibrated basis for selecting composition and porosity in NIPS-derived PI/h-BN packaging films.

CRediT ~~Authorship Contribution Statement~~authorship contribution statement

~~[editorial comment removed]~~[Name 1]: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft. ~~[editorial comment removed]~~[Name 2]: Resources, Supervision, Writing – review & editing, Funding acquisition. ~~Adjust roles as appropriate.~~[Adjust roles as appropriate.]

Declaration of ~~Competing Interest~~competing interest

The authors declare ~~no~~ that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data ~~Availability~~availability

All experimental data underlying the figures are provided in the Supplementary Material and in the accompanying raw-data workbook. The ~~complete~~ Python implementation of the ~~3-phase~~ three-phase Lewis–Nielsen model, including the parameter-estimation protocol, is listed in ~~Supplementary~~ Section S5 of the Supplementary Material and archived in a public repository (~~[editorial comment removed]~~**[DOI]**).

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